STUDENT RESEARCH · HPC & BRAIN CAVE

How HPC Supports AI Research for Thai Stocks

A controlled, verifiable process for comparing five computer-learning methods in a Thai-stock research setting.

503

planned experiment runs

5

AI learning algorithms

29.7%

verified official-plan progress

HPC makes a large research plan manageable, traceable and repeatable.

What Is the Computer Learning?

Deep reinforcement learning means practising decisions repeatedly and learning from feedback. The five names below are different learning styles.

PPO | A2C | DDPG | TD3 | SAC

1 · Observe

Read historical prices, trading volume and market context.

2 · Decide

Choose how much of the portfolio goes to each stock.

3 · Learn

Use feedback from returns, risk and transaction costs.

4 · Test

Evaluate on later dates that were not used for training.

The design uses repeated evidence, not one lucky result.

Why HPC Matters

A laptop is useful for small tests. A 503-run plan must be scheduled, recorded, checked and reproducible.

PERSONAL COMPUTER

- Build and debug code
- Run small trials
- Inspect results quickly

HIGH-PERFORMANCE COMPUTING

- Schedule computing tasks
- Record job IDs, runtime and resources
- Verify, recover and repeat work

HPC matters not only for speed, but for reliable research at scale.

A Real HPC Run

STORAGE-SAFE SMOKE TEST

```
# one controlled scheduler submission
job_id=$(sbatch --parsable smoke_test.sbatch)

# verified result
Job 3550
State: COMPLETED
Elapsed: 00:33:20
Algorithms: 5
```

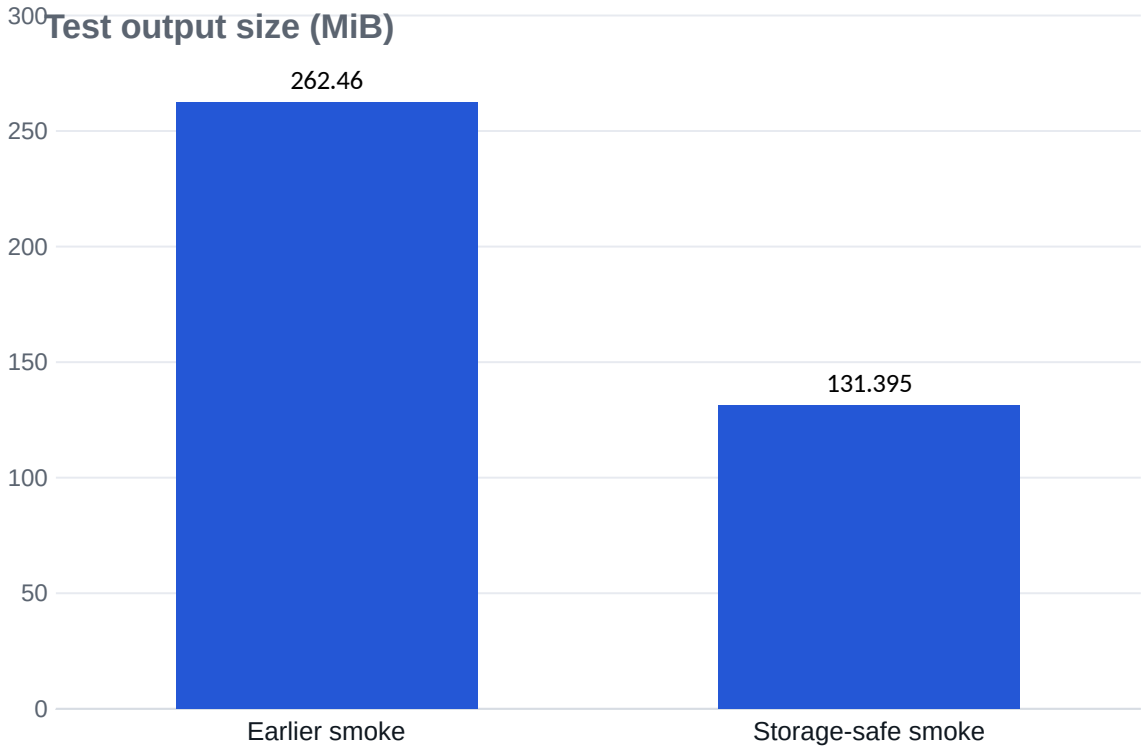
What is a smoke test?

A small trial that checks the complete process before a much larger experiment is submitted.

COMPLETED
33 minutes 20 seconds

Every large campaign begins with a small, identifiable and checkable run.

Responsible Use of Storage



MiB is a standard file-size unit.

49.94% less test output

5 + 5 experiment manifests + retention receipts

0 large intermediate checkpoints retained

The project nearly halved its test output while preserving the evidence needed for verification.

What Is Done - and What Is Not

VERIFIED

11 / 37

official project checkpoints complete (29.7%)

- Storage-safe HPC test completed
- Five algorithms produced complete records
- Exact runtime and resource evidence preserved

STILL AHEAD

150 trials

full screening has not been submitted

- Remaining checkpoints are incomplete
- Storage capacity must be checked again
- No final research conclusion yet

Ready for the next step is not the same as finished.

What Happens Next?

1

Recheck capacity

Confirm storage and computing resources before opening a new execution window.

2

Run 150-trial screening

Submit only when fresh safety checks pass, in one controlled HPC window.

3

Compare and preserve

Analyze all five methods and keep records that others can inspect or repeat.

HPC turns a large idea into research that can be measured, checked and repeated.